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Defining the intension and extension of nations' sovereignty in the age of generative AI

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ABSTRACT

Generative artificial intelligence's (GAI's) wide-ranging potential for generation, communication and dissemination of information marks an unprecedented level of transformation by challenging states with varying approaches to state sovereignty. These implications underscore the necessity of examining how states maintain control over digital territories that are being reshaped by the increasing influence of GAI. In particular, we explore how economic, political and social circumstances will shape government regulations on artificial intelligence (AI). This research highlights the importance of debate concerning how the notion of sovereignty itself may be shaped amidst the proliferation of GAI.

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Artificial intelligence; generative AI; sovereignty; digital territory

1. Introduction

An increasing number of social, political and economic activity takes place within digital territories, where governments are competing with other state and non-state actors for control. Current innovation that encompasses the emergence of generative artificial intelligence (GAI), wherein machines are capable of content creation in response to human-defined prompts, shifts AI from simply being relevant to national interests, to having the capacity to disrupt national sovereignty. The criticality of artificial intelligence (AI) is further evident based on the implications for information control and innovation that GAI potentially holds for the relationship between a government, private companies and civil society, all of which are key stakeholders connected to one another through the generation and dissemination of information within digital territories.

While technological advancement in a broader sense can be viewed as a backbone of sovereign power, nations have attributed its *meaning* to different *referent objects*, thus, complicating the relation between, respectively, *intension* and *extension* of sovereignty. With this in mind, we have identified the underlying research question for this manuscript as follows: How have different states adapted their sovereignty within the digital territory in response to the GAI race facilitated by mass production of AI-generated content? And how have the social, economic and political circumstances affected the formulation of government policies and regulations concerning the use of AI?

Our research question builds on current digital sovereignty research, such as Jiang (2024) who acknowledges that a state's digital sovereignty is not an objective concept; rather, a state's digital sovereignty will be shaped by their social, economic and political circumstances evidenced by decisions around investment, policy and regulations. For AI, sovereignty's link with a state's circumstances is uncertain due to the tension that arises with AI threatening and protecting sovereignty, similar to digital sovereignty and expansionism (Roberts et al., 2023).

While the intersection of AI and sovereignty is a growing point of interest globally, as will be demonstrated by examples in the empirical framework section, a case study method focusing on two influential

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jurisdictions is used in this research to identify multiple factors that connect GAI to sovereignty in two contrasting settings. Russia and China, referred to in this article as 'harder regulations' or 'HR states' following Fratini et al.'s (2024) typology, are grouped together as states that have adopted the information sovereignty approach. States in the 'softer regulations' following Fratini et al. (2024), also referred to as 'SR states' in the manuscript (European Union, US and United Kingdom), have a history of encouraging globalisation of digital spaces, and are, thus, grouped together. While these powers may also be classified as authoritarian and democratic regimes respectively, with regards to AI, the approaches taken by states may have fewer distinctions and, thus, should be seen as a spectrum rather than a framework with strict labelling. We chose to focus on influential jurisdictions in this article to highlight internal and external aspects of sovereignty as well as differences between their regulatory approaches.

Using historical analysis, we explain how advances in technology have impacted aspects of state sovereignty including its application within digital territories. As such aspects are often a result of social construct, conveyed by narratives and pursued through policies (Pohle et al., 2024) we adopt content analysis and narrative analysis methods within the case studies. We analyse available national-level regulatory documents such as Russia's National AI Strategy, the People's Republic of China's (PRC's) Interim Measures, the EU AI Act, the US Executive Order and the UK GAI Framework to explore the impact of GAI content on state sovereignty through (a) interpreting the relations between existing sovereignty attributes and GAI development and (b) establishing the reasons for these connections. The case studies' observations are limited to GAI models to provide an understanding of how the ability of both state and non-state actors to mass-produce digital objects by utilising GAI can impact state sovereignty and control over digital spaces. By accounting for important political, economic and social factors similar to Jiang's (2024) study, this research explores how the increase in speed, scope and quality of digital objects that are now mass-produced by utilising GAI algorithms redefines internal and external legitimacy.

Thus, the contribution of this research is three-fold: first, rather than AI governance in general, it focuses on the content produced by GAI, including the ability and capacity to produce such content, as an aspect of sovereignty. Secondly, we contribute to the discussion on digital sovereignty and reconceptualisation of territory by explaining how GAI can be utilised both to strengthen and threaten state sovereignty. Finally, we suggest that development of GAI cannot be viewed as a race for technological superiority on its own, but more of a result of social, economic and political circumstances of sovereign states.

This analysis will be presented in six sections, Section 2 presents a theoretical conceptualisation of how the notion of sovereignty combined with technological development has underpinned contemporary international politics and the emergence of digital territory. Section 3 develops these points by examining the empirical implications of how the increasing spread of AI applications can be perceived as both a threatening to and enabling control over digital territories reflected within internal and external legitimacy of state sovereignty. Sections 4 and 5 are two empirical case studies, investigating how countries' policies and actions regarding GAI are affected by internal and external attributes of sovereignty. Section 6 discusses the conclusions, that reflect on the broader impact of GAI technologies and the implications for international relations, security, industry and policy alignment, are then presented.

2. Theoretical framework: technology's implications for sovereignty

The concept of sovereignty, often associated with Machiavelli, Hobbes and Bodin, is often summarised as 'supreme authority within a territory' (Philpott, 2020). Sovereignty can also be viewed as a monopoly on violence where a state alone has the right to use or authorise the use of physical force (Weber, 1919), or logistics of power (Mann, 1986). It can be understood as a legal mechanism: 'The whole body of rights and attributes which a state possesses in its territory, to the exclusion of all other states, and also in its relation with other states' (Corfu Channel Case, 1949, p. 43), or in terms of a function: 'Sovereignty is the concept by means of which modern political and legal philosophy elaborates the problem of the contingent unity of a political community' (Lindahl, 2006). Stephen Krasner (1999), while challenging the normative side of the concept, argues that sovereignty has four distinct attributes including domestic (exclusive legal authority within a territory), interdependence (control over transnational flows), international legal sovereignty (recognition of states) and Westphalian sovereignty (principle of non-intervention). These definitions highlight internal and external aspects of sovereignty where internal legitimacy means

designation of all acts of jurisdiction and governance as a sole responsibility of the state, and external legitimacy corresponds to the principle of equality among the state and *only* (emphasis added) states (Prokhovnik, 1996, p. 8).

In the Westphalian system, state-making cannot be done without occupying a certain geographical space and effectively controlling it through adopting legislation and regulations (Glanville, 2013; Williams, 2010). The factors that have generally contributed to establishing sovereignty over specific territory were technological advances: cannon technology, for example, increased their range and correspondingly expanded the distance of domestic territorial waters that states claimed (Howland, 2019). Technological developments reached an unprecedented pace in the twentieth century, as illustrated by the US and Soviet space programmes, driven by a recognition that the ability to deliver rockets into space also enhanced the power projection capabilities of their nuclear warheads; furthermore, the ability to launch satellites into space also granted the superpowers enhanced capacity to project their ideological narratives. These technological developments fundamentally enhanced the ability of states to broadcast communications and new ideas beyond their borders, altering the understanding of sovereignty as being solely based on a government's control of its landmass.

With the increasing digitalisation, encryption has been viewed as a means to enhance 'data sovereignty', a right to control and manage one's own data, which, under the Westphalian system, is considered an extension of state sovereignty. Encryption is the process through which particular communications are transformed so that, even if they are intercepted, their meaning cannot be discerned (Buchanan, 2016). This allows the preservation of privileged access to 'sovereign data', even in the context of a borderless internet. Many countries have focused on developing policies and laws ensuring that the country's data remains in servers 'on shore', but the enforcement of such laws are hard to implement or scale. The most practical means to preserve sovereignty could be the preservation of the control of encryption keys and related cryptography-related export controls. This approach also adds another aspect to Norma Möllers's definition of 'digital territory' as 'nationalized information infrastructure' (2021).

Klimburg (2018) describes the cyberspace model as a four-layered inverted pyramid: in addition to the physical layer at the bottom (infrastructure and hardware), then neural (programmes and protocols) and data (information streams) layers he distinguishes a layer of social interaction – what Morris (2023) also calls 'digital spaces'. The national cyber infrastructure typically provides a foundation for digital spaces that cover the space beyond national borders; some states, however, see the information and interaction in the digital spaces as an object sovereign control. Combined with the difference in perspectives on sovereignty, this model can also explain why there is no consensus on the definition of sovereignty for the cyber domain: Couture and Toupin (2019) note that such terms as 'technological sovereignty', 'digital sovereignty', 'network sovereignty', 'data sovereignty' and 'information sovereignty', etc., are all applied to describe similar concepts related to one or more layers of digital territory. Within international law, territory itself can also be reconceptualised as 'a product of social relations' based on the contemporary governance, law-making and regulatory practices (Lythgoe, 2024), also reflecting the jurisdictional complexities with digital territories possibly overlapping each other.

Although AI is not a new concept – initial discussions can be traced to the 1950s (Calo, 2017) – recent proliferation of large language models (LLMs) have increased its importance. While the literature appears, at the time of writing, to be lacking an accepted definition of AI (Ashraf, 2020), governments typically view the phenomenon as an algorithm-based system that can predict, recommend or decide based on human-defined objectives (US Congress, s3[3]; UK Information Commission Office, 2023) though AI was initially conceptualised to replicate human cognition processes (Haigh, 2023) highlighted in Russia's definition of AI as being capable of simulating humans (Zysk, 2023). The shift from cognition processes to a more practical task completion may reflect AI's history, including its oscillations of growth, and states' belief that AI is key to digital sovereignty demonstrated through funding and resources from state and military, which initially AI development was dependent on. As the concept of sovereignty is often applied to important developments in technology, naturally, the discussions of 'AI sovereignty' (Mügge, 2024), and 'sovereign AI' have been prevalent in policy (Alduhishy, 2024) and business circles (Strier, 2024). The internal aspect of AI sovereignty can be conceptualised as AI infrastructure control, training data, and the content GAI produces with the aim of controlling the domestic population, while the external aspect reflects engaging in international regulation and/or competition. Alnahhas and Yousef (2024) note that GAI, as a flexible tool, can

generate output asserting a state's interests contributing to internal sovereignty. However, since digital territories include non-state actors such as private businesses and individuals, the external aspect is not limited to the state actor. This intersection of AI and sovereignty is further explained in the section below.

3. Empirical implications of AI on sovereignty

3.1. AI as a threat to information control and sovereignty

In facilitating the process through which users can enter prompts to rapidly generate data and content at a far higher pace, GAI has increasingly aroused concerns for countries over their capacity to assert and maintain state sovereignty over the digital spaces. These human-defined objectives and input, including the training and testing data, will be crucial in shaping what content is created. Yet, AI tools are typically developed by non-state actors. Thus, Calderaro and Blumfelde (2022) in their study on the European Union's ambition and approach to AI highlight that a state would likely need to cede control or data sovereignty over AI applications and generated information. With a limited ability to control the produced information, this may put state's interests and control over digital territory at risk especially if GAI-created content is potentially politically sensitive or contradictory to a state's interests. Building on Daskal (2015) who notes that the characteristics of data in the modern era, e.g., its mobility and independence from location, challenge typical territorial discourse around jurisdictions, such as what jurisdiction rules apply, enforcing state controls over territories can be similarly difficult for GAI, which can largely exist in digital territories. This adds a level of ambiguity to GAI and its application.

For instance, GAI enables creating multiple chatbots usable on social media networks for the purpose of spreading information narratives to gain control over digital spaces as demonstrated by misinformation campaigns in 2023 in Venezuela (Timmers, 2021). Such activity falls under the umbrella of hybrid warfare operations (Siman, 2024), conducted under the threshold of using conventional force in response, but at the same time undermining both internal (monopoly on control within a certain jurisdiction) and external (principle of non-interference) sovereignty of a state. Amplified by GAI capabilities to fabricate written, audio and visual content, a malicious actor presents a distorted image of its adversary, accompanied by AI-generated audio-visual content and text to provide a facsimile of evidence in support of the narrative, then proceeds to hijack the discourse in an attempt to undermine its adversary's authority in the digital territory (Lyngaas, 2023). Besides the detrimental impact on prior censorship efforts, the potential loss of information control affecting domestic population control can weaken a state's sovereignty. In instances where the state actively regulates information shared or distributed through censorship mechanisms to protect their sovereignty, AI may directly diminish that sovereignty. Such an impact has implications comparable to social media's role in information dissemination and social mobilisation against the censorship and authoritarian political control. GAI, if unregulated, can provide the same ability on a potentially larger scale. For instance, GAI tools can be used to create and provide information that may weaken state sovereignty and strategic narratives; concurrently, the ability of GAI tools to create content at a pace that fact-checkers, which may be manual or automated using AI tools with or without human oversight, cannot match could undermine the efforts at censoring such content.

Further, even when trained with millions of observations, LLM models often become prone to systemic or structural biases which could lead to material consequences of specific groups of people (Goode, 2021, p. 368). Independent studies have repeatedly shown that LLMs demonstrated cultural bias towards values similar to English-speaking and Protestant European countries (Tao et al., 2024), bias towards Western-associated aspects of culture (Naous et al., 2024) and better performance on cognitive psychological tasks resembling Western, educated, industrialised, rich and democratic (WEIRD) countries (Atari et al., 2023). As LLMs were also found to exhibit political (Bang et al., 2024; Urman & Makhortykh, 2023) and social (Dhamala et al., 2021) bias, they can be viewed as an exercise in privilege and an instrument of power by the big tech (Talat et al., 2022), as illustrated by instances of AI generating covertly racist decisions based on people's dialect or skin tones (Hofmann et al., 2024). Cases of unsuccessful AI use in public services in Austria for labour market opportunities, United Kingdom for risk-monitoring and the Netherlands for fraud detection in claiming child benefits have demonstrated the AI embedded discrimination (Timmers, 2021).

Thus, AI challenges both the internal and external aspects of sovereignty. On the one hand, the principle of all states – and *only* states – as being equal in international relations is challenged by large corporations and early winners of the AI race in having control over data used in training AI models and, by extension, digital territories. On the other hand, the extent to which AI-embedded biases intrude on fundamental rights has, in turn, led to a failure to deliver effective public services, like the Netherlands's technology to detect child support misuse, thereby undermining their internal legitimacy (Timmers, 2021). In light of the starkly differing digital capacities of various states, such a development portends the prospect of an information-based world of 'have' and 'have-not' states, and the corresponding implications of 'unequal' levels of sovereignty with large corporations having unprecedented control and influence over AI impacting state sovereignty unless regulated as suggested by the literature, e.g., Rikap et al. (2024) suggests that introducing taxation or other mechanisms may reduce the threat to sovereignty of large technology corporations.

3.2. AI as an innovation to enable sovereignty

Though the open nature of AI challenges traditional conceptions of sovereignty, there is potential for AI to strengthen sovereignty due to its relative affordability compared to typical censorship, monitoring and disinformation tools (Feldstein, 2021). States can then quickly deploy tools, such as AI bots to distort information without being constrained by typical barriers, such as the time and cost involved to mobilise physical forces (Stukal et al., 2022).

According to Feldstein (2021), there are two ways that states can use AI tools to strengthen their sovereignty over digital territories: to monitor discontent/maintain control or spread disinformation to manipulate legitimacy of internal or external opponents to their regime. Firstly, monitoring discontent occurs when the state will track specific keywords, hashtags and other meta-data that characterise patterns of discontent in their state (Feldstein, 2021). This pattern is fed into an AI algorithm to identify matching individuals. States may then undertake actions to maintain control, e.g., restrict the individual's information sharing powers, bearing similarities to traditional state sponsored censorship.

Secondly, AI tools may be used to spread disinformation. For instance, automated bots can be used on social media to influence what information is shared and propagated (Freedom House, 2023). This may include state-approved information or disinformation. An example is the use of AI-generated bots, including foreign state bots during electoral periods to influence voters, such as during the 2020 US Presidential Election (Ferrara et al., 2020). More recently, OpenAI also linked accounts using ChatGPT to generate content on the US presidential elections to an Iranian influence operation (OpenAI, 2024). Furthermore, GAI can generate deepfakes to discredit individuals, typically activists or those against a state, as in the case of Alexei Navalny with deepfake videos of his wife and his mother surfacing after his death in February 2024 (DeepMedia, 2024; Marchant de Abreu, 2024).

Additionally, deepfakes can be generated to present false narratives. One example is the AI-generated video demonstrating female support of the former prime minister of Pakistan, Imran Khan, rather than the military forces (Freedom House, 2023). Other examples include deepfake videos of the president of Ukraine Volodymyr Zelenskyy calling on his country's military to surrender after Russian invasion in 2022.

As a response to the potential power of AI tools, some states choose to territorialise their AI capability through investment, policy and regulation. For example, Singapore's most recent National AI Strategy commits significant investment (over SGD$$1 billion) over the next five years to build Singapore's AI capacity (Singapore Ministry of Digital Development and Information, 2024). Likewise, the European Union and India have made similar investment plans, €2.1 billion and US$$1.25 billion respectively (European Parliament, 2024; Ghosh, 2024).

Non-state actors have made similar efforts (Filgueiras, 2022). For instance, the International Organisation for Standardisation released ISO/IEC 42001:2023, an AI management system standard (2023). The Organisation for Economic Co-operation and Development (OECD) has also developed several tools and conducted significant research to enable the development of safe AI (2024).

Finally, GAI can be seen as a technology defining international competition among state actors for spheres of influence. China's DeepSeek, although praised for its accessibility and adaptability, was quickly banned on government devices in Australia, Taiwan and South Korea, with the latter suspending downloads

of DeepSeek country-wide after its data protection authority requested improvements to be made in accordance with the country's privacy law. ChatGPT, however, was earlier suggested to be used for parliament audits by the ruling party (Kim, 2024). This case reflects how GAI applications can become the centre of sovereignty concerns and control over digital territory largely due to the tension arising from GAI's ability to simultaneously weaken and strengthen sovereignty affecting how a state chooses to control and leverage GAI, acknowledged by Roberts et al. (2023), which is further illustrated in the next two sections.

4. Case study 1: Russia and China

Both China and Russia, as states with harder regulations compared to other states (Fratini et al., 2024), have sought to maintain tight control of their digital territory in a bid to utilise cyberspace for economic as well as military and intelligence purposes, whilst simultaneously seeking to prevent cyberspace from being used by domestic political dissidents and foreign influences to organise opposition to the ruling parties. At the Fourth Plenum of the Chinese Communist Party (CCP) in 2004, then-Chinese President Hu Jintao proposed a 'Harmonious Society' – interpreted in the West as a euphemism for state censorship – for the purpose of 'maintaining social stability' (China Media Project, 2021). As early as 1998, China's Ministry of Public Security had introduced the Golden Shield Project to control the flow of information into China and block the flow of sensitive foreign content. Since 2012, the CCP has unveiled increasing levels of efforts to control the media narratives surrounding China's domestic circumstances as well as China's relations with the international community. Domestically, this has taken the form of blocking the websites of foreign media, social networking sites and other potential rallying calls for dissent against the CCP, whilst supporting accounts utilise various channels including US-based YouTube, Facebook and X, to circulate CCP approved ideologies.

Similarly, according to Orlova (2020, p. 230), the Russian state has 'framed the internet as dangerous, unreliable and biased' to their citizens, thereby justifying state censorship and restrictions. Embedded in existing, amended or new legislation, these restrictions can inhibit Russian citizens from accessing or posting information contrary to the Russian government narrative. Some examples of relevant legislation include the Internet Restriction Bill in 2012 detailing blocked websites, Anti-LGBT Propaganda Law, Law on Pre-Trial Blocking of Websites, Anti-Terrorist Laws and Russian Information Act (Allerson, 2022; Polatin-Reuben & Wright, 2014). In particular, the Law on Pre-Trial Blocking of Websites allows for rapid blocking of any websites that may have content contrary to the Russian government narrative (Allerson, 2022).

In addition, there is active monitoring of social media by the Russian state. For instance, Russian social media users may face criminal charges if their social media posts are considered to constitute a threat to Russian values and their narrative (Allerson, 2022). More recently, according to a law proposed in 2024, owners of social media channels that gained more than 10,000 subscribers must report their personal details to be verified by Roskomnadzor (federal monitoring service) which maintains a public list of trusted social media (TASS, 2024).

4.1. AI in Russia and China

Regarding AI, Russia and China have taken a similar approach as they did for the Internet: treating it as a tool to be controlled and managed, in line with their views of information flow control and security as a part of their *internal* sovereignty mirroring their physical control efforts in a digital territory context. Both states have invested significantly in AI innovation and applications to monitor information flow: China has invested USD 7.76 billion in 2023, and Russia planned to allocate 224 billion Rubles by 2024 (equivalent to approximately USD 2.5 billion making it the fourth highest globally if included in the AI Index Report) (Markotkin & Chernenko, 2020; Stanford University, 2024). China's investment, while significantly less than the US (USD 67.22 billion), is still the second highest investment by a country.

The level of investment is testament to the importance that both states place on AI. For instance, the criticality of Russia becoming an AI power is characterised by this statement from President Putin (The Economist, 2024): 'whoever leads in AI will rule the world'. Similarly, China has declared its ambition to become a world leader in AI by 2030 (Mozur, 2017). The extent to which China and Russia have invested

in AI is reflective of their beliefs that GAI and the content it generates can strengthen their power or has the potential to challenge their sovereignty and weaken their control over digital territory, as indicated by public state statements on the contribution of AI to global power, e.g., President Putin stated that becoming a leader of AI will help a country become the 'ruler of the world' (as cited by Petrella et al., 2021).

Aware of the importance of controlling the information flow within China's cyberspace as an internal aspect of China's sovereignty, the CCP has invested heavily on AI generally (Beraja et al., 2024), including enhancing the regime's tools for internal surveillance of the populace for any signs of dissent towards the regime. This includes AI applications that notify the authorities when Chinese internet users use Baidu to search for information concerning information the CCP deems sensitive. In previous decades, the users had devised means of circumnavigating state censorship through homophones, the creative use of wordplay and other indirect references for the purpose of criticising the CCP: for instance, using the term 朝廷 (*Cháotíng*, or 'Royal Court') as a reference to increasingly centralised power, and 笑贫 (*Xiaopin*, or 'laughing at the poor') to refer to Deng Xiaoping (Han, 2018, p. 87). Whilst this approach provided a measure of leeway to criticise the regime, the ability of AI to enhance state surveillance through data mining for such coded keywords has sharply reduced the loopholes available. Herein, the knowledge to Chinese internet users over the extent to which they face comprehensive state surveillance on cyberspace functions as an instrument of deterrence through which the CCP can make credible threats of punishment to dissuade the masses from using cyberspace to communicate potentially controversial content (Pei, 2024). In Russia's case, the significant costs of the Ukraine war have led to an increased focus on AI to demonstrate their power while developing their military capacity into intellectualised warfare (Bendett, 2024). Though unspecified, their rationale for focusing on AI may be its relative affordability and scalability compared to other methods used to establish control over digital spaces (Edmonds et al., 2021).

Despite the level of investment by China and Russia, the extent of their progress in becoming AI powers varies. Both states have developed their own alternatives to ChatGPT, e.g., GigaChat and YandexGPT-2 from Russia, and ErnieBot and Doubao from China (The Economist, 2024; Yang, 2023). In addition, these models are monitored to ensure they are continuing to produce narrative-fitting statements. For example, two Chinese AI bots (Xiao Bing and BabyQ) were taken down and reprogrammed after they began responding with statements contrary to the state narrative (Xu, 2018). At the same time Russia's domestic GAI applications are scrutinised internally for their inability to produce answers in accordance with the Russia's government stance to war in Ukraine. Ex-president Medvedev and the Human Rights Council chief Fadeyev blamed Yandex's Alisa and VKontakte's Marusia for their refusal to answer the questions on Donetsk and Luhansk regions status with Fadeyev admitting that even ChatGPT produces more satisfactory answers (Medvedev, 2024; The Moscow Times, 2024).

China's Qwen 2 on the other hand, when receiving a prompt in Mandarin Chinese that could be considered sensitive by the Chinese government typically answers along the lines of the government's official narrative, however, when the same questions are posed in English, the model produces significantly more refusals to answer (Lin, 2024). This contrast in responses dependent on language of query may indicate that the majority of the data enabling Qwen 2 is sourced from China likely leveraging China's centralised internet system and their efforts towards centralising their data (The Economist, 2018; Wong & Dobson, 2019). DeepSeek's answers also align with the state narrative affecting responses (Yang, 2025).

The research and development of AI varies in the two states. China led AI patents globally in 2022 with expectations that this will continue (Stanford University, 2024). In contrast, little is known about Russia's AI research and development though President Vladmir Putin released an updated AI strategy (Faulconbridge, 2023; The Economist, 2024). This limited progress and knowledge about the progress is to be expected considering the consequences of the Ukraine war, which has inhibited Russia's AI capability. One consequence is Western states putting pressure on other states to limit the access Russia has to components and data, e.g., GitHub removing access for Russian software developers (Bendett, 2024). The ability to participate in AI innovations in other states is also limited due to sanctions (Bendett, 2024). Another consequence is the loss of skilled Russian professionals that occurred due to the war (Davis, 2022), although the expertise of information and communications technology (ICT) professionals has been considered Russia's main strength in AI. Russia's AI strategy also points out the lack of computing power as the main challenge to stable AI development (Art. 1716). Hence, Russia's ability to leverage AI power in control over digital territory is potentially constrained.

Besides their investment in AI tools, both states are active in presenting a controlled narrative around foreign AI tools. For example, Russia positions the AI developments of other states as a threat to its internal sovereignty (Nadibaidze & Miotto, 2023). Russia's AI Strategy states that one-sided constraints imposed by adversary states (designated as 'unfriendly states' in official Russian documents) in 2022–2023 have posed challenges to the domestic AI industry resulting in a lack of computing power, human resources, low level of AI integration in public administration, lack of subsidies, regulatory barriers, information security and data privacy threats, unfair competition, limiting the access to new technologies and international cooperation. More specifically, recent economic changes due to the international sanctions could lead to AI-specific information security threats (Art. 1716), such as data poisoning (NIST, 2024). The strategy estimates that failure to match the leading AI powers would result in Russia's economic and technological inferiority (Art. 18).

In comparison, China's Interim Measures for the Management of Generative Artificial Intelligence Services released in 2023 directly address the risks GAI could pose to its sovereignty by stating that GAI services should, among other things, comply with the Core Socialist Values, and not produce content 'inciting subversion of state power, overthrowing the socialist system, endangering national security and interests, harming the country's image, inciting separatism […]' (Art. 4.1). To achieve successful GAI governance, China encourages 'the adoption of safe and reliable chips, software, tools, computational power, and data resources' (Art. 6) and targets an increase in computing power by more than 50% by 2025 in response to the US export controls (Ye, 2023).

Reflecting the *external sovereignty*, Russia and China have invested in AI to provide a tool that could challenge other actors' control over digital territory, such as through heightened military capability and spreading disinformation (Watts, 2024). This latter example, spreading disinformation, is already occurring. For instance, Russia used an AI software package (Meliorator) to generate fake online personas to spread disinformation, pro-Russian and anti-Ukraine information online (Harding, 2024). Similarly, China is reported to have used GAI to spread disinformation, such as generating news and social media content relating to the Taiwanese elections including an AI-generated recording suggesting that a Taiwanese election candidate, Terry Gou, had endorsed another candidate (Koh, 2024).

AI being positioned as a form of global competition against the democratic states (Nadibaidze & Miotto, 2023) is more noticeable in Russia's rhetoric. Russia's AI strategy envisages global competition between a handful of leading AI powerhouse countries attempting to ensure their continuous dominance and economic benefits in the AI market by imposing constraints to prevent other states from entering (Art. 16). While most of the competition is attributed to economic development, GAI models are said to pose 'a threat of dissemination of prohibited information, violation of copyright, and generation of wrongful facts', highlighting the importance of control over digital territories (Art. 174). Although facing international restrictions and sanctions in AI research and development, Russia is seeking global influence in AI regulation and ethics by establishing an AI Ethics Code and having partner states adopt it. International cooperation is still designated as an important priority of AI development in Russia, including cooperation at various international fora such as the United Nations and G20. It is, however, notable that Russia has concentrated its outreach in AI collaboration through the BRICS Outreach and BRICS+ forums, along with the Collective Security Treaty Organisation (CSTO) and Shanghai Cooperation Organisation (SCO) (Art. 5113).

While facing the restrictions accessing GAI technology such as ChatGPT (OpenAI has restricted the use of its applications in Russia due to security concerns), Yandex search engine returns links to paid and unpaid services that grant access to ChatGPT without a virtual private network (VPN) or registration as top searches. So, as a contrast to Russian government banning some Western applications such as Twitter and Facebook with VPN services also being constantly suspended, illegal access to the services restricted from use in Russia by the Western companies is provided and even encouraged despite the possibility of the LLMs generating results that the Russian government would be wary of.

However, Russia's 2024 AI strategy highlights the need to pursue cooperation with the leading international AI research centres and acknowledgement in the international academic community (Art. 5113). Success in domestic GAI developments is also often framed in the context of the West: while T-Bank AI Research group has developed the software called ReBased that is capable of processing long texts, it acknowledged the role of Stanford University in developing the original model and underlined

the importance of international academic cooperation in the LLM research (RIA News, 2024). Positioning GAI success within the international competition and cooperation frameworks also reflects a facet of external legitimacy with actors pursuing the status of equality among others. In addition, China is heavily involved in global AI collaboration and governance through organising a United Nations workshop on AI and their support for a United Nations resolution on AI safety and managing the digital divide across countries (Mishra, 2024).

5. Case study 2: softer regulation states (European Union, US and United Kingdom)

The European Union (EU), US and United Kingdom (UK) have a relatively softer approach to protecting their sovereignty and digital territories compared to China and Russia's approach. Potentially, this softer approach may reflect the acceptance of the trans-jurisdictional context of GAI making typical approaches to protecting territories less effective. The SR states' approach is apparent in more flexible laws and governance (Flonk et al., 2024), permitting free rein over market mechanisms, and protection of freedom of speech. For example, the US has emphasised 'promoting innovation … letting market mechanisms rule' (Stanger et al., 2024, p. 451). During the mid-1990s, amidst the increasing commercial spread of cyberspace, the US, having hegemonic control of emerging technology, focused on gaining economic benefits from promoting its tech companies. Madeline Carr notes that the process of Internet commercialisation was intensely promoted by the Clinton-Gore administration, as it was seen as an engine for economic growth necessary to ensure continued dominance in the international system (Carr, 2016). The US did not try to establish any guidance for AI regulation until 2019, given the extent to which the US sought to maximise its economic power by exporting AI technology through the dominance of its technological companies. Previously the DC District Court upheld the decision for TikTok to divest from its Chinese parent ByteDance by January 2025 or face a full ban in the US having based its decision on the need to protect the freedom of speech 'from a foreign adversary nation and to limit that adversary's ability to gather data on people in the United States' (United States Court of Appeals, 2024). Likewise, in September 2024, the US Government announced plans to ban Chinese and Russian technology in US vehicles (Sherman, 2024).

Prior to 2018, the EU had a similar approach to the US. However, since 2018, stronger legislation has been implemented, such as Regulation 2021/784 on the Dissemination of Terrorist Content Online. Flonk et al. (2024) notes that this shift from soft to harder governance mirrors the increased prioritisation of safeguarding digital territories considering the growing threats to democracy posed by misinformation and conspiracy theories online. The growing geopolitical tensions globally may also encourage a firmer protection of digital territories. Yet, the European Union's AI Act restricts predictive profiling applications, treating them as high-risk applications (Art. 5(1) (d)). Typically, predictive profiling is used in counter-terrorism efforts (Briend, 2025), relevant to sovereignty as terrorism is positioned as a threat, reflecting the softer approach taken by the EU.

Besides protecting themselves from threats, SR states view safeguarding their digital sovereignty and territories as critical for protecting their strategic autonomy through reducing dependence on other states, particularly 'enemy' states, relating to technology and cyberspace. Specifically, the EU defines digital sovereignty as 'Europe's ability to act independently in the digital world' (European Parliament, 2019, p. 1).

5.1. AI in SR states

With this shift to a harder form of governance over digital territories, AI is considered a critical tool for SR states to govern and protect their sovereignty and digital territories reflected in their semi-soft approach to AI. Typically, the SR states tend to focus on the external aspects of sovereignty through innovation, and international competition and cooperation to safeguard their digital territories and control other territories. In addition, SR states maintain *internal aspects of sovereignty* through efforts to regulate AI developers, which while not censorship will enable SR states some control over what is developed and deployed in their territory. The scrutiny and investigations into AI developers, such as investigations by the European Union's Data Protection Board (2024) into ChatGPT/OpenAI, provides an indication that some SR states may maintain internal sovereignty through seeking evidence from AI developers of compliance with internal sovereignty expectations, e.g., not transferring personal data outside of the EU.

Regarding innovation, SR states will typically invest or support investment in AI to enable ongoing innovation as an attempt to territorialise AI infrastructure. For example, the US has the highest level of investment in AI (The Economist, 2024). Such efforts may be driven by the perceived contribution of AI to military capabilities by states (Calderaro & Blumfelde, 2022), similar to Russia. However, much of the AI investment in SR states tends to be from non-state actors reflecting a shift where non-state actors are now concerned with digital territories as well as the state. For example, the scale of investment from private companies has allowed the AI lobbying to nationalise the discussion on AI regulations, in particular, its opposition to California's AI bill (vetoed by the governor) despite employees of OpenAI, Google's DeepMind, Meta and Anthropic sending a statement in support of the bill. Slowing down innovation is usually a major point of criticism for any attempt at AI regulation both from AI companies and politicians (Davies, 2024). AI regulation enacted in California, where 35 out of world's top 50 AI companies are domiciled, would provide a foundation for alignment of AI regulations worldwide, however, placing restrictions on domestic businesses is seen as potentially reducing their financial profit, a key factor for economic security in the US. Thus, enabling AI development is considered crucial for internal legitimacy for the US.

As much of the AI development occurs by non-state actors, SR states have tended to focus predominantly on the ethics of AI including ensuring safety for society. These efforts tend to focus on regulating AI developers and organisations using AI though these efforts are balanced against innovation as evidenced in the US by the above described lobbying by non-state actors and the revoking of the Executive Order for select AI companies to commit towards developing responsible AI (US White House, 2023). Typically, this focus is demonstrated through regulatory actions, such as the EU's recently passed AI Act (European Commission, 2024) for developers and deployers of AI controlling what they can do in the EU. This Act highlights the systemic risks that general-purpose AI models could pose including, among others, 'any actual or reasonably foreseeable negative effects on democratic processes, public and economic security; the dissemination of illegal, false, or discriminatory content' (Art. 110). Besides the European Union's legislation, similar regulatory actions are less common in other SR states, at the time of writing, though most SR states have released strategic documents illustrating their planned AI pathway such as UK's GAI framework for HM Government. This difference in focus is reflective of the US and the UK being more market-oriented than the EU with regard to digital sovereignty (Fratini et al., 2024).

Similar to Russia's view of *external sovereignty*, SR states also focus on the idea of AI as a competition (Stix & Maas, 2021). For instance, the US's discourse on AI emphasises that the US should 'lead' AI development, notable in President Biden's Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence (US White House, 2023). In addition, the US is concerned about 'losing this [AI] race to China' as it may lead to significant consequences for the US (US House Armed Services Committee, 2020, p. 1). As part of leading AI innovation, the US is also concerned with territorialising the physical infrastructure enabling AI performance, i.e., computing power as an increase in computing power was found to improve performance of LLMs accordingly (Kaplan et al., 2020). As AI development requires a large amount of computer memory, data storage and the most advanced processing units, the US enacted the CHIPS and Science Act in 2022 with $$52.7 billion in subsidies for US semiconductor manufacturers followed by President Trump's current intention to improve energy supply (Volcovici & Renshaw, 2025) while imposing export bans on the advanced chip technology to China. The hypothesis is that successfully restricting China's access to AI chips would significantly stall its accumulation of computing power and give the US an advantage over China in several aspects of the AI race from building automated weapons systems to running GAI applications. Furthermore, the strategy of 'onshoring' is seen as necessary to make the US more competitive in all stages of semiconductor production by reducing the dependence on other countries (in particular, Taiwan and South Korea), thus, guaranteeing self-reliance in the development of AI technology as well.

6. Conclusion

This article positions the development of GAI based on the intension and extension of state sovereignty within digital territory. While using its established definition ('supreme authority within a territory'), the application of the sovereignty concept to advanced technologies was shown to vary based on states' interpretation of it. Different approaches to AI sovereignty mean that GAI policies could be adapted to

address different facets and potential applications within different digital spaces. Policy alignment on advanced technologies, including AI, could contribute to improved privacy regulation, private sector accountability, cybersecurity and international security. The empirical framework provides an understanding of how AI can be viewed simultaneously both as a threat and a tool to advance both internal and external legitimacy of the same actors. Although generative AI has created opportunities for a wide range of actors to undermine internal and external aspects of state sovereignty, states have also demonstrated their willingness to develop and territorialise GAI as a tool for process streamlining, monitoring and mobilisation through investment, policy and regulation.

Two case studies showcasing contrasting approaches to sovereignty of influential jurisdictions provided the analysis of GAI and its use based on political and social circumstances and the states' technological and financial resources and capabilities. In these case studies the countries were grouped according to their different views on the scope and application of state sovereignty to digital territories and GAI. On the one hand, China and Russia tend to mimic their place-based boundaries in the digital space for GAI where possible though China's approach is closer in alignment to the other countries studied in this article. In contrast, UK, US and EU appear to more readily recognise that GAI is a trans-jurisdiction issue meaning their focus is on securing their digital territory. As empirical evidence suggests, countries see generative AI as an affordable tool to propagate their narratives. At the same time, the authoritarian states are concerned about the capability of generative AI to disrupt information flow control, which is considered as one of the sovereignty pillars and is essential for the stable control over digital spaces. However, China and Russia's narratives slightly diverge on the drivers for generative AI development: while China emphasises adherence to Core Socialist Values, Russia's narrative is largely shaped by overcoming 'one-sided' geopolitical constraints. The scope of resources, both human and financial, available to the two nations also explains this difference in their approach to GAI as a tool for control of digital territories.

As in the case of traditional sovereignty, securing enough resources – in particular, by acquiring computing power necessary for training AI models – is something that both the one-party dominant states and the transatlantic region nations authoritarian and democratic states prioritise in their strategies as the backbone of fast and stable GAI. Competition in GAI development is also addressed by both groups of states, although, framed differently: while Russia's approach is largely shaped by the country's confrontation with the West, the US views winning the AI race in terms of its economic benefits for its tech companies. Innovation and regulation of AI are viewed as essential components of sovereignty by the countries of the transatlantic region. Although non-state actors such as private companies, that often drive technological innovation, might lobby against increasing regulation of generative AI, the ethics of AI use remains an important aspect in sovereign development of AI in the transatlantic region with the EU AI Act highlighting the potential dangers to democracy and economic security and, thus, most policies and regulations focus on AI developers and organisations using AI.

Despite the differences between the case studies, there is a critical and novel similarity across them relating to noticeable engagement with international policy alignment efforts. Traditionally, international policy alignment is not considered to benefit sovereignty, due to the context of dysfunctional international mechanisms and the absence of an international regulatory body. However, the case studies illustrate, irrespective of stances on GAI and its impact on sovereignty, that the mechanism of engaging with international alignment efforts is considered critical for protecting sovereignty from the threats of AI. Hence, China and the US efforts in the United Nations and Russia's similarly, but to a lesser extent, efforts in BRICS Outreach and BRICS+ forums relating to GAI. Similarly, we expect that an equivalent approach would be used for other issues arising in the cyberspace that spill over to neighbouring digital territories.

Largely, the emerging benefit of international alignment engagement is due to the specific nature of the GAI environment in the context of sovereignty. Specifically, the nature of cyberspace sovereignty within occupied digital territories, the nature of GAI as a rapid content-creation tool and the interdependencies between nations, due to the structure of the current semiconductor industry, are leading to states perceiving benefit in participating in international collaborations. Hence, maintaining sovereignty in the presence of GAI cannot be achieved without engaging with other states and further strengthening external legitimacy. A state needs to be an active participant and leader in efforts to establish global infrastructure and governance regarding AI. By leading these efforts, states direct the conversation to limit the impact of adversarial GAI use and protect their own interests. For example, China's efforts in the United Nation with GAI workshop

organising and resolution support is interpreted as positioning and aligning themselves with other states which provides some form of protection against those criticising their actions regarding GAI.

However, international collaborations can be fraught with tensions. For example, the development of high bandwidth memory chips (HBM) capable of increasing computing power benefitting GAI has led to the US imposing export restrictions on China relating to advanced memory chips (Chang & de Acosta, 2024) and Nvidia shifting some outsourced production to the US (Almeida, 2025). Other states are also heavily investing in HBM to enable their GAI competitiveness, e.g., South Korea's SK Hynix, a memory chip maker and a lead manufacturer of HBM, will invest US$$74.6 billion through 2028 for the development of chips to empower their GAI efforts (Asia ITNews, 2024). While the analysis in this article was limited to great powers, further research is needed to study how middle powers adapt their sovereignty to developments in GAI.

The pursuit of AI sovereignty has implications for many global and domestic industries, including semiconductors, traditional and social media, energy infrastructure, and national security. Because AI sovereignty encompasses a combination of technological tools and the diversity of their applications, GAI is highly dependent on, and at the same time has an impact on, other industries outside of the immediate digital territory. Investment in AI development requires policy support, a technological base, human resources and geopolitical issues. First, the semiconductor industry, in particular, is proving vital, as states with a strong semiconductor sector can use access to computing power to develop their AI sovereignty. On the other hand, policymakers are free to introduce restrictions on semiconductor technology that can limit the strengthening of their adversaries' AI sector. Second, the potential of GAI to mass produce content has already been impacting people's engagement with information and media globally (UNESCO, 2024). To counter growing risks, updating the media literacy framework as well as expanding fact-checking initiatives and building alternative GAI models trained on verified data could improve the media environment. Third, while AI can support energy distribution by integrating renewables, training LLMs also leads to a significant increase in power consumption and load on electric grids (Bourzac, 2024). This implies fortification of power systems and energy generation is necessary to sustain the development of AI which, however, could lead to depletion of energy and water resources and lead to the destabilisation of more traditional attributes of state sovereignty. Lastly, though there is little evidence of whether the military or government directly influenced AI research, the presence of funding indicates that the military and/or government perceived AI as an innovation capable of strengthening national interests. Perception of GAI as an innovation supporting national interests, namely sovereignty, has also expanded to recognising AI as a threat to sovereignty.

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